



NASA SPACE APPS CHALLENGE 2026 · TEAM PROTIKRITI

Martian Map.

A Google-Maps-style atlas of Mars for the first human explorers. It has a 3D globe, rover Street View, live weather, walkable terrain and a Marswalk planner that keeps astronauts off dangerous slopes. Everything is built from open NASA, USGS and IAU data.

2,087

rover Street View stops
(703 Perseverance · 1,384
Curiosity)

2,052

IAU-named places
searchable on the globe

30

Perseverance rock samples
at their exact sites

15°

maximum slope on every
planned Marswalk leg

Eight tools, one map of Mars

Every layer, stop and number comes from a published NASA, USGS or IAU dataset, with the source cited in the repository.

01 Whole-planet 3D globe

Spin and zoom around all of Mars on real topography, with high-resolution terrain where the rovers are.

- Global MOLA elevation; Jezero at 20 m and Gale at 32 m
- Search 2,052 named features, 16 lander sites and 30 candidate human Exploration Zones
- Shareable links that reopen the exact view

Data: MGS MOLA · USGS CTX and MSL DEMs · IAU Gazetteer · Viking MDIM

03 Safe Marswalk planner

Click a start and science stops; each leg is the fastest route that never crosses ground steeper than 15°.

- A* routing with Tobler's walking-speed model, off the main thread
- Distance, climb and descent, steepest step and EVA time
- Slope hazards hatched in red, never shown by colour alone

Data: USGS CTX DEM (Jezero) · MSL HiRISE DEM mosaic (Gale)

05 Live Mars weather

Today's conditions from the rovers' own weather stations.

- Curiosity REMS (live): air and ground temperature, pressure, UV, sky
- Perseverance MEDA (latest archive)
- Sunrise and sunset in local Mars time, and a Mars-year temperature chart

Data: NASA/JPL REMS and MEDA feeds, with snapshot fallback

07 Settlement site reports

Right-click anywhere on Mars for a one-page briefing for settlers.

- Elevation and terrain detail, daylight hours and season
- Likelihood of water ice in the top metre of ground
- Surface radiation, and the nearest named feature, landing site and Exploration Zone

Data: SWIM 2.0 ice · MSL RAD · MOLA · IAU Gazetteer · Mars24

02 Rover Street View

Stand where Perseverance or Curiosity stopped and look around in 360°.

- Each stop's Navcam sweep stitched live into one seamless sphere
- Walk arrows to the next stop and a compass heading
- Links back to every source frame on NASA's site

Data: NASA raw-image APIs (Navcam) · JPL MMGIS waypoints

04 Explore on foot

Drop onto the terrain at true scale and walk it like a video game.

- Keyboard walking, strafing and turning; drag to look
- Live position, heading and elevation
- Starts where the rover is today

Data: site DEMs draped with HiRISE and CTX imagery

06 Mission replay and samples

Play both rovers' journeys sol by sol, and visit every Perseverance sample core.

- Timeline scrubber with follow-camera
- 3D Perseverance model at its position on any sol
- 30 sample cards with rock type and sealing sol, linked to Street View

Data: MMGIS traverses · NASA Mars Rock Samples · NASA 3D Resources

08 Mars clock and coordinates

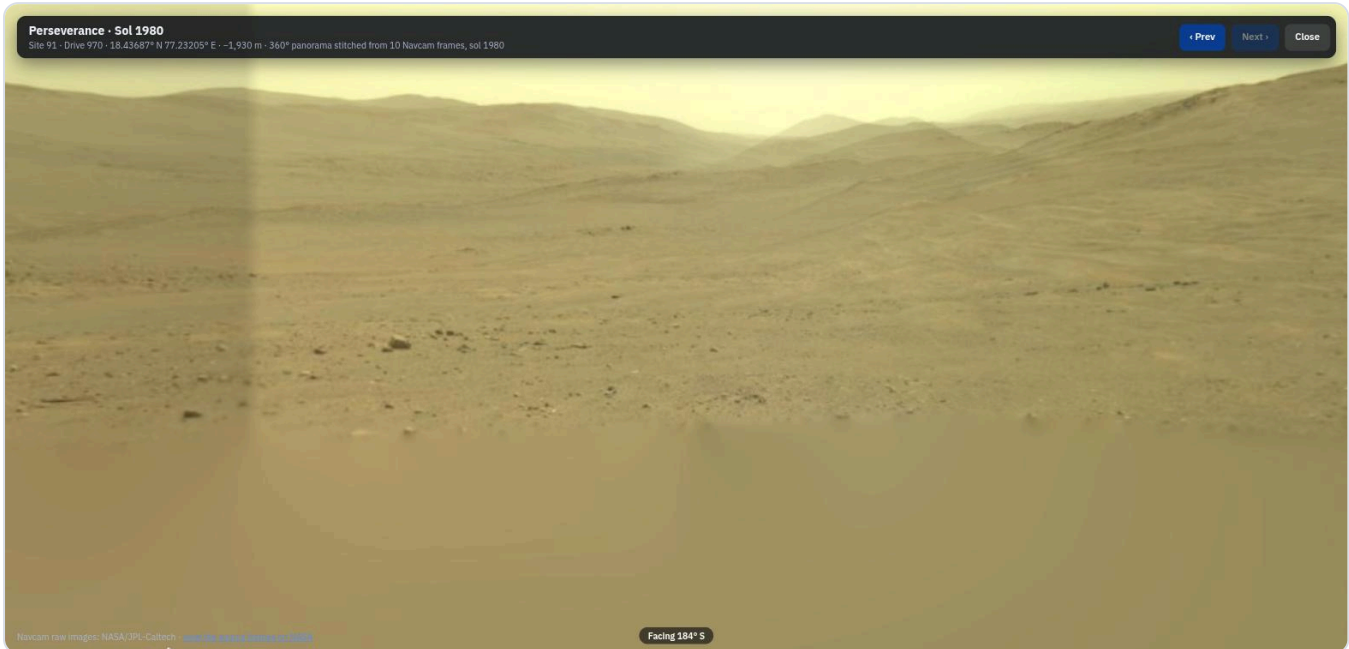
The time and place, measured the Mars way.

- Local solar time, sols for each rover, season (Ls) and Mars Sol Date
- Real sun position lights the globe; step through hours and sols
- IAU Mars 2000 planetocentric coordinates (Mars has no GPS)

Model: NASA GISS Mars24 algorithm · IAU 2015 Mars frame

Street View on another planet

NASA publishes rover photos one frame at a time. Martian Map turns each stop's frames into a single sphere you can look around.



Perseverance, sol 1980, Jezero Crater: 10 Navcam frames stitched into a 360° view, facing south.



Curiosity, sol 5016, Gale Crater: 7 frames stitched into a 219° view, with the rover's own deck in front.

1 · Pick the sweep

From all frames at a stop, keep the sequence that covers the widest arc and drop repeated shots.

2 · Place each frame

Project it through the camera's real pointing and the rover's heading onto a sphere.

3 · Match and blend

Correct lens darkening, balance exposure between frames, feather every seam.

4 · Fill the gaps

Blend unseen sky and ground smoothly from the nearby imagery. No black holes.

Built on real NASA data

Positions use the IAU Mars 2000 sphere (radius 3,396,190 m). Every derived file is rebuilt from source by one command.

DATASET	MISSION / INSTRUMENT	USED FOR
Global topography	Mars Global Surveyor · MOLA MEGDR	Globe terrain, elevation readout, colour relief, roughness
Jezero elevation	MRO · CTX DEM, 20 m (USGS)	Terrain, slope hazards, route planning
Gale elevation	MRO · HiRISE-derived MSL DEM mosaic, read at 32 m (USGS)	Terrain, slope hazards, route planning
Surface imagery	MRO HiRISE 25 cm · CTX · Viking MDIM (NASA Trek)	Photo-real ground at every zoom
Rover positions	JPL MMGIS traverses and waypoints	Tracks, Street View stops, mission replay
Rover photos	Perseverance and Curiosity Navcam raw images	Street View panoramas
Weather	Curiosity REMS · Perseverance MEDA	Weather panels and Mars-year charts
Buried ice	SWIM 2.0 (Subsurface Water Ice Mapping)	Settlement site reports
Radiation	Curiosity · RAD (Hassler et al. 2014)	Surface dose in site reports
Dust	Mars Global Surveyor · TES dust index	Dust-cover layer
Place names	IAU / USGS Gazetteer of Planetary Nomenclature	2,052 searchable features
Human Exploration Zones	NASA 2015 First Landing Sites workshop	30 candidate zones on the globe
Rock samples	NASA Mars Rock Samples page	30 sample cards at their sealing sites
Rover model	NASA 3D Resources · Perseverance glTF	3D rover in mission replay

Engineering quality

- 182 automated tests: 40 Python and 142 TypeScript
- Strict checks on every push: ruff, mypy --strict, ESLint, tsc, Prettier
- Routing and panorama stitching run in Web Workers, so the map never freezes
- Keyboard-operable, labelled controls; hazards never shown by colour alone
- Open-source stack: Python pipeline, TypeScript + CesiumJS web app

Honest limits

- Walking speed is an Earth model (Tobler), not tuned for spacesuits yet
- Slope maps resolve 20–32 m; boulders need 1 m HiRISE checks
- Routes are planned only inside the two high-resolution sites
- Perseverance weather is the latest archived data, not live
- Stitched Street View needs the included image proxy; plain static hosting shows separate photos